ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT

MATHEMATICS
Higher 2

9740 / 02

Paper 2

21 August 2009

JC 2 PRELIMINARY EXAMINATION

Time allowed**: 3 hours**

Additional Materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Index number, Form Class, graphic and/or scientific calculator model/s on the cover page.
Write your Index number and full name on all the work you hand in.
Write in dark blue or black pen on your answer scripts.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in the question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of **6** printed pages.

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ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT
JC2 Preliminary Examination 2009

MATHEMATICS 9740
Higher 2
Paper 2

/ 100

Index No: | | | |

Form Class: ___________

Name: ___________________________

Calculator model: ___________________

Arrange your answers in the same numerical order.

Place this cover sheet on top of them and tie them together with the string provided.

Question no.	Marks
1	
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13	

Section A: Pure Mathematics [40 marks]

1 The coordinates of the points A and B are (1, 2, 1) and (4, -2, 2) respectively.

Find the vector equation of the straight line AB. **[1]**

If C is a point on the line AB such that OC bisects the angle AOB, find the position vector of the point C. **[4]**

Find the ratio AC:CB. **[2]**

2 It is given that the first 3 terms in the expansion of $\left(2-ax\right)^{-b}$ in ascending powers of x are $\frac{1}{8}+\frac{9}{16}x+cx^2+\ldots$, where a, b and c are positive constants.

(i) Find the values of a, b and c. **[7]**

(ii) Show that the coefficient of x^r is in the form $p\left(r+1\right)\left(r+2\right)q^r$, where p and q are constants to be determined. **[4]**

3 The curve C is defined parametrically by $x=t^2$ and $y=\ln t$, $t>0$. The region bounded by the curve C, the x-axis, the vertical lines $x=2$ and $x=5$ is denoted by R.

(i) Sketch the graph of C for $2\le x\le 5$, indicating clearly the coordinates of the end-points. **[2]**

(ii) Express the area of R as an integral in t and show that it has an exact value given by $\alpha\ln 5+\beta\ln 2+\gamma$, where α, β and γ are constants to be determined. **[5]**

(iii) The region enclosed by the curve C, the y-axis, the horizontal lines $y=\frac{1}{2}\ln 2$ and $y=\frac{1}{2}\ln 5$ is denoted by S. Find the volume of the solid generated when S is rotated through 2π radians about the x-axis. **[3]**

4 (a) It is given that $z=1+i$ and $w=\sqrt{3}+i$. Without the use of a calculator, find the exact values of $\left|\frac{z}{w}\right|$ and $\arg\left(\frac{z}{w}\right)$. By expressing $\frac{z}{w}$ in the form $x+iy$, show that $\tan\frac{\pi}{12}=2-\sqrt{3}$. **[6]**

(b) On a single Argand diagram, sketch the loci given by (i) $\left|z-2-i\right|=2$ and (ii) $\arg\left(z+i\right)=\frac{\pi}{4}$.

Hence, or otherwise, find the exact values of all complex numbers z that satisfy both (i) and (ii). **[6]**

[Turn Over

Section B: Statistics [60 marks]

5 A family of five wishes to go for lunch at their favourite restaurant. They hopped into their family car which has two front seats and enough space for three persons behind. Find the number of seating arrangements in the car if only the father can drive. **[1]**

At the restaurant, the family was shown to a round table with five seats. Find the number of ways the family members could seat themselves if the youngest child insisted on sitting with her mother. **[2]**

If the family was asked to sit at a round table with six seats which were numbered from 1 to 6 instead, find how many ways the family members could seat themselves if the youngest child still insisted on sitting with her mother. **[3]**

6 Vehicles arrive randomly at a junction at an average rate of 10 per minute. Traffic counts at this location indicate that 70 percent of these vehicles are cars. Assume that the arrivals of the vehicle follow a Poisson distribution.

(i) Calculate the probability that exactly 18 vehicles arrive in a 2-minute interval. **[1]**

(ii) Find the variance of the number of cars that arrive in a 3-hour period. **[1]**

(iii) Using a suitable approximation, find the probability that there will be at least 30 cars in the next 50 vehicle arrivals. **[3]**

7 (a) A school consisting of 600 students has 400 boys and 200 girls. The principal wants to find out what students think of the meals provided in the canteen. On one particular day she selects a sample of 12 pupils. Explain briefly whether a stratified sample might be preferable to a random sample in this context. **[2]**

(b) In order to carry out a statistical test on the leaves of a tree, a student examines 150 leaves that are within reach. Comment on the validity of the sampling method chosen in this context. **[1]**

8 Two friends, Wayne and Henry played 120 matches of tennis. Wayne won 70 of them, while Henry only won 50. Assuming that they are equally skilled at tennis, find the probability that Wayne would win at least 70 matches. Comment on the assumption based on your answer. **[3]**

9 The average salary of fresh engineering graduates is \$20 000 per year with a standard deviation \$2 200.

(i) A placement firm conducts a phone survey of 50 randomly selected fresh engineering graduates. Calculate the probability that the average salary for this sample is within \$800 of the population mean. **[3]**

(ii) The placement firm will ultimately obtain five different and independent selected samples, each consisting of 50 fresh engineering graduates. Calculate the probability that three out of the five samples will have sample mean within \$800 of the population mean. State one assumption made in this calculation. **[3]**

10 Peter Piper the Pickle Packer provides a written guarantee with each bottle of his pickles that every pickle will be at least 8 cm long. His bottles are made such that the longest pickle that they can hold is 10.5 cm. The lengths of the pickles are normally distributed with mean of 9 cm and standard deviation 0.8 cm.

(i) Find the probability that a randomly chosen pickle Peter Piper picked will have to be discarded because

(a) it does not meet his guaranteed standard; **[1]**

(b) it does not fit the bottle. **[1]**

(ii) Peter wishes to discard no more than 5 percent of the pickles he picks. Comment on the validity of this statement **[1]**

(iii) Using a suitable approximation, find the probability that out of 100 pickles that Peter picks more than 8 will not fit the bottle. **[3]**

(iv) Find the probability that if Peter randomly picks two pickles, both pickles will fail to meet his guaranteed standard given that their total length is less than 16 cm. **[3]**

11 A machine produces ball-bearings whose diameters X are normally distributed. The mean of a random sample of n values of X is denoted by $\overline{X}$ and it is given that $\mathrm{E}(\overline{X}) = 3.6$ mm and $\mathrm{Var}(\overline{X}) = \dfrac{\sigma^2}{n}$ mm^2. After modification of the machine, a random sample of n bearings was taken and the mean of their diameters was found to be 3.5 mm.

(i) State a null hypothesis and an alternative hypothesis that the manufacturer may use to test whether the modification has decreased the mean diameter of the ball bearings. **[1]**

(ii) Using a 1% significance level, calculate the least number of ball-bearings taken which would lead to the rejection of the null hypothesis assuming that the population standard deviation σ is 1.2 mm. **[4]**

(iii) A second random sample of 15 ball bearings was taken and the diameters, x mm, of each ball bearing is measured, with result summarised by

$$\sum x = 47 \text{ and } \sum (x-1)^2 = 90.$$

Carry out another test at the 5% significance level, with the same null and alternative hypotheses as above without assuming that the population standard deviation σ is 1.2 mm. **[4]**

State the meaning of p-value in the context of the question. **[1]**

[Turn Over

12 The moisture content, m, of core samples of mud from a lake is measured as a percentage. It is believed that m varies with the depth x metres at which the core is collected. To test this, samples were taken at different depths, and the results are shown in the table below.

x	0	5	10	15	20	25	30	35
m	90	82	56	42	30	21	21	18

(i) Calculate r, the linear product moment correlation coefficient content for these data. State an implication of the value that you obtained in the context of the question. Comment on what this value implies about the two regression lines of m on x and that of x on m. **[3]**

(ii) Use a suitable line of regression to estimate the depth when the moisture content is 50%. Comment on the reliability of the estimate. **[3]**

(iii) The pH value, y, of the mud also depends on the depth x metres at which the core is collected. To test this, the pH value, y, of the above sample was also taken. It is given that the linear product moment correlation coefficient between y and x is -0.904 and the least square regression line of y on x is $y = -2.33x + 50$. Find the equation of the least squares regression line of x on y. **[3]**

13 A street ball match between Singapore United and Singapore Rovers ends in a draw and the match is to be decided on the result of penalty kicks. One round of penalty kicks consists of two kicks, one kick being taken by each team. If both teams score or if both teams fail to score, then another round of penalty kicks is played. The match is won when, in a round of penalty kicks, one team scores and the other team fails to score.
The probability that Singapore United scores a goal with a penalty kick is 0.8 and independently, the probability that Singapore Rovers scores a goal with a penalty kick is 0.9.

Find the probability that the result of the match is still undecided after 1 round. **[2]**

(i) Find the probability that Singapore United wins the match in less than 3 rounds of penalty kicks given that Singapore United scores a goal in the first round of penalty kicks. **[3]**

(ii) Find the least number of rounds of penalty kicks n such that the probability that the result of the match is decided in at most n rounds of penalty kicks is greater than 0.98. **[4]**

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